QUANTT Credit Trading

Project introduction and open roles · 2026/27

Simon Jarvis, team lead · simon.jarvis0@gmail.com
github.com/quanttqueensu/CreditTrading

The project

We operate a systematic credit strategy on a $500,000 Interactive Brokers paper account. The capital is simulated; the prices, orders, fills and trading costs are real.

The system runs unattended. Each weekday it refreshes its data, assesses whether trading is safe, transmits its orders, records the executions, and raises an alert on failure. This year we intend to keep it running, establish whether it survives real trading costs, and research the next strategy.

The strategy

A closed-end fund trades on an exchange but issues its shares once, after which the share count is fixed. It publishes the value of its holdings daily as the NAV, while the share price is set independently by the exchange. The two diverge.

In an ordinary ETF they do not, because large banks can exchange shares for the underlying bonds and back, trading against any gap until it closes. We measured this: the gap on high yield ETFs compressed from 188 basis points in 2008 to 3.8 by 2026, and that compression killed our first two strategies. Closed-end funds have no equivalent mechanism, so the gap opens and persists.

Our strategy scores each fund's gap against that fund's own history over the prior year, holds the unusually cheap funds long and the unusually rich funds short in equal dollar amounts, and is therefore indifferent to market direction.

We spent much of the summer attempting to show this was a conventional exposure in disguise. Against high yield, investment grade, interest rates, equity and volatility, those five factors explain half a percent of the returns. The result also held against the closed-end fund sector's own returns, the strongest control we could construct.

Current position

The signal is established. Trading costs remain the open risk.

The strategy must decide in the evening, because its signal requires the NAV, which publishes after the close. On the first live day the orders rested overnight and filled at 07:27, two hours before the exchange opened, when these funds have negligible depth. We paid 0.94% to trade against a modelled 0.10%. Priced at decision prices the book was up $50; priced at fills it was down $7,350.

We moved to orders that execute in the closing auction, the deepest liquidity of the day, but have not tested that change because the broker platform has been down since August. Restoring it is our first priority.

One further correction should be stated before anyone joins. Our backtest assumed an entry price that is not obtainable, and correcting it reduces the expected Sharpe ratio from 0.82 to approximately 0.51. We would rather recruit against the accurate figure.

New members do not start from nothing. The summer produced a daily feed of prices for 11,423 bonds built from public filings at no cost, five scheduled jobs that run the book unattended, and a written record of twelve strategy ideas that failed, with the cause of each.

Roles

No finance background is required. Our documentation covers the finance from first principles.

Quant Researcher, 2 to 4 positions. Testing new strategy ideas and establishing whether they are real. Most are not, and reaching that conclusion quickly is the core skill; promising results are attacked before they are reported. Requires Python and pandas, basic statistics including regressions and t-tests, and the discipline to abandon a preferred idea when it fails a test.

Execution and Infrastructure, 1 to 2 positions. Ownership of the broker connection, the schedulers, order flow, and the measurement of realised trading cost. This area has produced more insight than any backtest, because live systems fail in ways research code does not expose. Requires solid Python, patience with silent failures, and ideally exposure to APIs or scheduled jobs.

Risk and Reporting, 1 to 2 positions. Weekly comparison of live performance against backtest expectation across returns, exposures, drawdowns and slippage. Requires basic Python and the ability to produce a concise written summary the team will read.

Goals and timeline

PERIOD	OBJECTIVE
September	Recruit and onboard. Backtest reproduced on every machine; broker connection restored.
Oct – Nov	Accumulate 60 live sessions, then execute the review committed to in writing before launch. The strategy is shut down if it fails.
Dec – Jan	Pre-register the January trade, then run it. Positive in 20 of the past 23 Januaries and uncorrelated with the main strategy.
Feb – Apr	Continued research, a resizing decision based on live evidence, and the final report and handover.

Applications

Send a brief note stating the role of interest and any relevant work to simon.jarvis0@gmail.com. The repository is public. `HOW_WE_GOT_HERE.md` is the recommended starting point and assumes no finance background.